

CE 527 : Advanced Pavement Engineering

Course Code : CE 527

Course Name : Advanced Pavement Engineering

L-P-T-C: 3-0-2-4

Intended for: UG and PG

Prerequisites: On approval of the course instructor.

Mutual Exclusion: NA

Approval : 57th BoA

Course Contents

- **Cement and Concrete Pavement Technology** (18 hours)
 - **Cement and Concrete Pavement Technology:** Cement Production and Cement Chemistry, Effect of mineral and chemical admixtures (including grinding aids). Supplementary cementitious materials and its effect on hydration Properties of hydrated Portland cement Fresh paste structure and properties; Concrete Pavement Fundamentals: Concrete pavement types, Elements (typical) of concrete pavement types, evolution of concrete pavement design, types of loads on concrete pavements, typical response of concrete pavements to load, typical distresses in concrete pavements. (12 hours)
 - **Pavement Design Methods:** Empirical approaches to the design of concrete pavement, Mechanistic - Empirical approaches to design of concrete pavements, Portland Cement Association, IRC:58-2015, Function of joints, Types of joints, Dowel bars vs tie bars. (6 hours)
- **Bituminous Technology and Analysis & Design of Flexible Pavement** (11 hours)
 - **Bituminous Technology:** Physical and chemical characterization of bitumen. Aging of bitumen, Binder properties and their relationship to pavement performance. Modification of bitumen, modified binders such as polymers and rubbers. (6 hours)
 - **Analysis & Design of Flexible Pavement:** Stresses and Deflections in Homogeneous Masses Elastic Layer Theories - Wheel Load Stresses – VDF. IRC 37: 2018 (5 hours)
- **Pavements Maintenance and Evaluation** (8 hours)
 - **Pavement Surface Condition & Its Evaluation:** Various Aspects of Surface and their Importance; Skid resistance, Pavement Condition Index (PCI), International Roughness Index (IRI), Causes, Factors Affecting, Measurement of skid resistance, Pavement functional and structural evaluation and its importance. (4 hours)
 - **Pavement Overlays & Design:** Pavement Overlays, Design of Flexible Overlay over Flexible Pavement by Benkelman Beam Deflection and other Methods, Flexible Overlays and Rigid Overlays over Rigid Pavements, Use of Geosynthetics in Pavement Overlays. (4 hours)
- **Recent trends in Pavement Technology** (5 Hours)
 - **Guidelines and practices:** Cold mix technologies and warm mix technologies: materials, additives, guidelines and practices. Perpetual Pavement, White topping, roller compacted concrete pavements, interlocking paving blocks, pervious concrete pavements, precast concrete pavements for highways and airfield; industrial pavements; concrete pavements for low volume road. (5 Hours)

Laboratory Experiments: (Total no of tests: 10)

1. Pavement Materials Testing,
2. Bitumen Content of RAP,
3. Rotational Viscometer Test,
4. Evaluates the viscoelastic properties of bitumen binder under different stress and temperature conditions,
5. Aging Tests on bitumen,
6. Rapid Chloride Penetration Test (RCPT),
7. Benkelman Beam Deflection test,
8. Heat of Hydration test,
9. Compressive strength test for cube test specimens
10. Flexural strength tests for beam samples

Text books:

1. H.F.W. Taylor, **Cement Chemistry**, 2nd Edition, Thomas Telford, 1997
2. Yang H. Huang, **Pavement Analysis and Design**, 2nd Edition, Pearson, 2004

References:

1. Shaw, D. J., **Introduction to Colloid and Surface Chemistry**, 4th Edition, Butterworth, 1992.
2. Hewlett, P. C., **Chemistry of Cement and Concrete**, 4th Edition, Elsevier Science & Technology Books, 2004.
3. Delatte, N., **Concrete Pavement: Design, Construction and Performance**, Taylor & Francis, 2008.
4. Riveiro, B., and M. Solla, **Non-Destructive Techniques for the Evaluation of Structures and Infrastructure**, CRC Press, 2016.
5. Odler, I., **Special Inorganic Cements**, E&FN Spon (Taylor & Francis Group), 2000.
6. Relevant AASHTO Guidelines.
7. Relevant ASTM Testing and Methodologies.